Robin Wigglesworth on Hyperscalers' 1.5 Trillion of Off-Balance Sheet Liabilities, Private Credit, and His Book "A Fabulous Debt"

Podcast: Monetary matters

AI Executive Summary

1. Core Thesis The current AI and data center boom represents a historic shift in fixed income financing, characterized by massive capital expenditure driven largely by off-balance-sheet leverage (leases and guarantees) totaling over $1.5 trillion among hyperscalers like Google and Microsoft. While this borrowing model mirrors transformative industrial manias of the 19th century (e.g., railways and canals), it introduces significant systemic risk if treated as "safe." Robin Wigglesworth argues that financial markets are prone to treating risky structures as safe, a pattern that culminated in the 2008 crisis and is now recurring in private credit. The future of this cycle depends on whether these technological investments mature sustainably or collapse under their own leverage, with the bond market's role as a transparent mechanism being increasingly overshadowed by opaque corporate borrowing games.

2. Key Arguments & Data

3. **Hyperscaler Leverage:** Googleâ€™s commitment to buy commitments is approximately $800 billion (including chips, memory, equipment, cooling, and power). Combined with lease obligations, hyperscalers' off-balance sheet leverage has grown from roughly $1 trillion last year to approximately $1.5 trillion currently.
4. **Off-Balance Sheet Structures:** Companies like Meta are using joint ventures with developers (e.g., Blue Owl) where they guarantee future lease payments for 20 years. While not always showing as traditional debt on balance sheets, these obligations are binding and often include "payment in kind" mechanisms to defer principal repayment.
5. **Private Credit Expansion:** The private credit market has absorbed a massive influx of capital post-2018, with over half of the market now rated double-B. Despite this perceived expansion, default rates may be higher than backward-looking data suggests due to refinancing cycles and unrealistic recovery rate assumptions (often assuming 70-80% recovery).

6. **Debt Cycles & Safety:** The 2008 Financial Crisis was distinct because institutions treated risky debt as "safe" money (e.g., AAA-rated MBS), whereas typical capital raising in the past has involved accepting higher risk/reward. The current cycle risks repeating this error with AI compute assets that may not appreciate or depreciate as expected (a Michael Burry-style depreciation thesis).
7. **Historical Parallels:** Wigglesworth draws direct comparisons to the 1873 Railway Mania and the US State Canal Bonds of the 1840s, where $10 trillion worth of bonds were issued during "manias." These resulted in catastrophic state bankruptcies and defaults (Jay Cooke's collapse), transforming economic infrastructure despite the initial financial pain.
8. **NVIDIA & Compute Finance:** A deal between NVIDIA and asset managers (Blackstone, BlackRock, KKR) to finance chips involves roughly $500 billion, structured as 80% debt and 20% equity. There is significant uncertainty regarding the long-term value retention of GPUs compared to infrastructure like oil pipelines.
9. **Rating Agency Endurance:** Despite AI advancements in analyzing credit prospectuses, rating agencies (S&P, Moody's, Fitch) remain durable because humans require "shorthand" models for risk assessment. They are enshrined in law (e.g., insurance asset requirements), though there is growing concern regarding "private label" ratings from insured private equity firms that may be inflated to maintain leverage capacity.

10. **OpenAI & Revenue:** OpenAI and Anthropic's financial health is debated; while Anthropic appears more healthy, revenue figures often include mark-to-market valuations of their AI investments rather than pure cash flow, leading to "non-real" earnings reports.

11. Actionable Takeaways

12. **Scrutinize Hidden Liabilities:** Investors should look beyond headline bond issuance numbers and examine "purchase commitments," lease guarantees, and capital expenditure projections that may appear off-balance-sheet but bind companies for future payments (e.g., Google's 800 billion purchase commitment).

13. **Treat Debt as a Cycle:** Expect volatility in AI-related debt assets. If technology transforms the world, the end-of-cycle bust could be painful, similar to railway manias; therefore, leverage limits should be respected, particularly for "safe" infrastructure assets that are priced like risky tech investments.
14. **Avoid Private Credit Complexity:** Retail investors should avoid illiquid private credit or BDC structures that promise flexibility without corresponding lockups. Complex financial engineering (like funding agreements backed notes) creates systemic opacity that increases risk rather than safety.
15. **Evaluate Rating Agencies Cautiously:** While rating agencies remain useful shorthand tools, they are not infallible. Investors should be skeptical of "Investment Grade" private credit loans, particularly those issued by insurers or firms with ties to private equity, as these may lack true transparency.
16. **Distinguish Equity from Debt in AI:** Recognize the difference between transformative tech cycles funded by equity (often self-sustaining) versus those funded primarily by debt (riskier if tech underperforms). The NVIDIA/BlackRock deal demonstrates a shift towards debt-heavy financing for hardware, increasing systemic risk.
17. **Prepare for Infrastructure Bust:** As CapEx bubbles have historically burst over 18 months or more, the current $1.5 trillion commitment level implies that liquidity may eventually tighten as hyperscalers seek to refinance or repay existing obligations, potentially leading to a correction in private credit returns.

Full Transcript

I'm joined today by Robin Wigglesworth, editor of AlphaVille, the Financial Times financial blog and author of A Fabulous Debt, the epic story of how bonds built the modern world. Robin, we want to talk about bonds, fixed income, of course, but we've got to start. with what's going on right now. You've been doing some work on the off balance sheet, hidden leverage of the hyperscalers, Meta, Google, Microsoft. So we're reading, oh, 100 billion of CapEx, Google's doing that, Microsoft's doing 150 billion. This is so much money. Oh my God. But actually it's looking like it's almost guaranteed to be way, way higher. You've been looking at the numbers and just from the first to the second quarter, the guarantees, the lease obligations and so forth off balance sheet again.

went from roughly one trillion to roughly 1.5 trillion. What are you, what are you looking at? No, I mean, it's fascinating. I mean, it is one of the biggest capital markets events of, you know, our lifetimes, really. It's just, I mean, we've seen massive CapEx booms before, like the railways in the 19th century is like the classic parallel that people draw, transformative technology, very expensive to build. What's unusual, of course, you know, railways back in the day used to be almost like venture capital ideas, right? They were very sort of VCE-ish. Today, it's like major large money machines that are doing it. So Google Alphabet makes lots of money, Meta makes lots of money, Amazon makes lots of money. And for a long time, the money they were pouring into data centers, you know, they could just fund it from their free cash flow. You know, Google search, Amazon, Facebook itself, just, you know, prints money, so it's easy.

But the scale is just becoming so massive that they've increasingly turned to the debt markets. As we now see, there are actually some signs of indigestion, like the sheer scale of the bond sales. We're talking multiple hundreds of billions of dollars, both last year and already this year. We've already smashed last year's record for the hyperscalers bond sales. And they're getting more creative. And look. Creativity in finance can be a good thing. I find a lot of this stuff fascinating, but it can also be quite dangerous as you know, Jack. And, you know, first it was structuring some of the

bonds as leases. So essentially, let's say, take a great example, Met is building a huge data center in Louisiana called Hyperion. And rather than do some pay squillions to build it, they're essentially sort of...

They're only investing 20%, they're buying 20% of it, but they are guaranteeing that they will lease that data center for 20 years. So, and the lease payments are essentially will cover the cost of that company itself, like a JV with Blue Owl and they'll sell those bonds to other investors. But, you know, so it's off balance sheet. It's not doesn't come up as as a as a bond or a debt or loan or for matter. But of course, it's on the hook for paying this lease for 20 years. And this, you know, has inspired a lot of the other hyperscalers. So we've seen massive amounts of these kind of lease structures happen. So that's what's gone to one point five trillion dollars. I mean, up from, I mean, less than trillion last year and nothing de minimis a few years ago.

And, you know, crucially, you know, 500 billion or so of that, you can see as sort of the leases have started so you can see them in the financial accounts. They won't appear as debt, but you'll see the payment obligations on the balance sheets. But a trillion dollars of that is for leases that haven't even started yet. And that doesn't appear except as a footnote. So Goldman Sachs, that's where I got the numbers from. They did the... God's working going through all the filings to find their stuff. What I did then was I started looking at the purchase commitments because these companies have also promised to buy obviously chips and equipment, cooling stuff like that, but obviously power as well. You know, these data centers need electricity and they need quite often guarantees that they will get that power. And that has gone from, again, also roughly 1 trillion earlier this year to 1.5 trillion.

And these are quite often payment obligations they can't squirrel out of. So they kind of walk, talk and quack a bit like debt, but they don't actually pair as debt. And I think it's fascinating. Maybe this is, you know, they're betting the house on AI and I hope this all works out. But I'd feel maybe slightly more comfortable if they structured some of this is more plain vanilla debt and let the debt markets do the talking. structured it as plain vanilla debt, we, you know, Goldman and you wouldn't have to do all of this work to figure out. It's, it is really interesting. So it's really a fine transformation

from a pure, complete triple A double A investment grade counterparty like Microsoft. And it is using that, but through a much less investment grade data center developer or a neocloud.

to then they're the ones who are actually spending the money and they report to investors this giant backlog, which we've just reported. And that giant backlog is basically these off balance sheet commitments that the Microsofts and the Hyperscalers have made. Yeah, I mean, you can say, you know, this is entirely disclosed. I mean, these companies aren't hiding the fact that they're pouring hundreds of billions of dollars into CapEx. And that money has to come from somewhere. They're not hiding this. But I do think they could Perhaps have chosen a slightly more transparent approach to this rather than wanting to preserve the optics of pristine balance sheets They are kind of increasingly getting creative by like how they structure how they pay how they dispose it just literally going through the ten cues to look at the purchase commitments, you know some of these companies Google for example actually was admirably like they don't break it up But then you can search for it and find it fairly easily on some of the other companies I had to spend, you know, quite a lot of time digging it out. It's not easy. And I think that's unfortunate. Who's the most complicated? I guess all the others accept Google. Some of them disclose it, but they don't disclose it in a uniform way. They use different words in different places. So it's hard to compare one quarter to the next quarter. And some of them, you know, don't disclose it at all, really. They just say they have, you know, material upcoming payments or promises to make payments. So it's a bit of a mixed bag, really. But Google, maybe because it's the biggest. I mean, Google has purchased commitments now of 800 billion. So that's chips, memory, equipment, cooling, electricity, the whole January. But that's almost half the total. I think maybe their accountants maybe got a little bit worried and thought, you know, we need to be quite transparent about this. And do you have a rough sense of.

over how many years the majority of that is scheduled for because like if that's scheduled for the next three years, then the Google CapEx number for expectations is too low. So Google using them again, they broke it out a little bit more transparently around 200 billion of that 800 billion is what they call short term. They don't define it there, but I'm pretty sure that'll be

over the next 12 months or so. So their payments are coming due. On the leases, you know, if the leases haven't started yet, it's an accounting thing. Like once a lease is started, you actually can put a right of use asset on your balance sheet and then put the liability on the other side. So it's just transparent. This isn't then necessary doing something bad or changing anything or doing anything differently from how people have done it forever. I just think that the scale of it is now such.

that it's a whole new world and maybe these off-balance sheet liabilities because they are financial liabilities that are in many cases extremely hard to square all out of. Methods for guarantee, for example, of the leases for the Hyperion data center are incredibly strong. I don't see how they can square all out of them. Then they should be more transparently disclosed and then investors can make their minds up and... Broadly speaking, I think people are OK and understand that they are spending this money. These companies are hiding it. But I wish some of the gamesmanship could maybe be cut out and them breaking out a little bit more, obviously. Robin, what do you make of the.

deal that the member of understanding that NVIDIA made with a five or six giant alternative asset management firms Blackstone, BlackRock, KKR to finance chips and recognize that these are an investable asset class. I was looking, I asked Claude before this, what percentage of that 500 billion is going to be debt versus equity? They said roughly 80% debt, 20% equity. So you're the debt guy. So this is good. We're speaking to you. I don't have this this is a debt cycle. It's not I mean There's lots of hoopla around like the IPOs or SpaceX and anthropocon and opening out to come But really this is a debt cycle That actually makes me more worried. I mean fundamentally like Incredible transformative technologies come around occasionally and sometimes they come true the railways were genuinely transformative as were canals or telegraph balls or the internet But when these sort of big investment cycles are primarily equity finance you know, they can break bad, but it's generally fine. I mean the end of the dot-com bubble, the stock market dropped 50% Picotroff. Economically, it was a nothing burger. It's kind of hardly disentangled from the effects of 9-11. That fueled capex cycles are very different. Like even when, you know, the under-minuteline premise comes true and AI kind of transforms the world, quite often they

end in tears. On this specific deal, one thing I would say that, you know, it's very easy.

to push out press releases saying we're going to lend or invest X or Y or Z into this or that I will see what actually materializes. I mean clearly there is a lot of heat in this area now and everybody wants to be seen to be leaning into it but it's going to be really interesting to see what they actually do and how they structure it because I think a lot of these investment firms in particular are going to be very careful about how they protect their own balance sheets but also their balance sheets of their investors. So I urge people at this point in the cycle to take, you know, press releases with a pinch, maybe even a fistful of salt. Yes. What do you think that these AI securities are going to look like? I mean, Larry Fink literally said, I don't know if he's on vacation, these AI securities, what's an AI security?

Well, I think it's just compute. And I think that's quite an interesting thing. And I think this story is kind of throwing a lot of different things. But it's the transformation of compute, like just kind of you can buy a Lisa certain amount of GPUs, I guess, or how you'd structure it. That can. can be turned into an asset class. I think it's, you know, there is a journey there, just because you say something is an asset class doesn't make it so. When it does become so, the SEC typically has something to say and we'll have a view about it. But I do think that is the direction of travel, that in the same way that like water or commodity is a tradable asset class, in least the futures on these things, I can see us getting compute futures as well. And that becoming roughly investable. My my weirdness is that you know just because something is investable last class doesn't mean people should be investing in it. We'll hear lots of stuff about oh you need to kind of be diversified and this is democratizing access to whatever everybody has and you know to paraphrase you know who was it.

gobbles, the Nazi propagandist who said, when I hear people talk about culture, I reach for my gun. He was not a nice man. But whenever I hear people talk about democratization of something, I tend to reach for my metaphorical gun because it's usually a code word for jamming something down the necks of retail investors that are not really quite ready to digest. So I actually have great hopes for compute futures and then turning

compute into some sort of tradable asset class that sounds cool and interesting to be. But I think there's a journey still to make on that. And I'd worry about what happens along the way as well. And something like a commercial real estate building. Yeah, it can be risky, but like, you know, a building is generally going to hold its value and generally appreciate over time. So like a loan to value of.

50% so so the you know you lend 50% of what the property is worth is like You know pretty pretty appropriate and you're lending to pipelines of oil infrastructure like these things these are things we understand and for something like compute Yes, the market right now is super hot and you know Google is buying compute from SpaceX for a super super high amount of money and on the depreciation argument like the Michael Burry argument basically every single data point of the past nine months has has not supported the Michael Burry depreciation point like depreciate like, you know, Cory just reported and they had, you know, for like six year old chips that they're going to lease out to 2029 at favorite favorable rates. But but just because Michael Burry has been wrong doesn't mean that like, there's going to be a glut at some time and that lending a trillion dollars against this thing is a good idea.

Well, fundamentally, it's a lending decision if you're lending towards it. I mean, the computer, the chips, they do degenerate. There is like, there is a half-life to how long you can keep them and, you know, for example, with SpaceX putting up in space, like, how are you going to do maintenance? How are you going to replace chips that are burnt out? But I would say that, look, you know, finance lending officers, like, they mess this up all the time. But broadly speaking, that's what markets are really good at. That can be priced. As long as it's a known risk, and you know, this is very much a known risk, people can press it in, people can adjust and sometimes they'll make a mistake and, you know, they'll get all their faces ripped off and other people will make money, but that's the way of markets. It's kind of what makes this system so dynamic. So, you know, I'd be worried about the sheer amount of money going in there and some of the return expectations and this kind of sense of, of FOMO that seems to be everywhere these days.

But that doesn't mean that the end destination isn't the right one. It's just, you know, how much money we lose along the way and who loses it and when. Robin, in your book, you've been doing a lot of work on, I mean, close to a millennia of reading of financial history. I'm curious about the trends and patterns you've noticed about the following question, pricing power. Whenever a new industry emerges, There often is tremendous pricing power, as there is in AI and semiconductors right now. Throughout your many, many centuries of reading of history, what tends to happen to that pricing power? It tends to erode. There are so many examples of this. I do worry, for example, right now, Nvidia is the picks in the shovels.

distributed to the entire revolution but it has obviously it depends on its own supply chain and I do wonder about the the assumption that nobody else can create GPUs at scale and quality of an Nvidia ever because right now we're pricing that in and pricing power tends not to last forever it's just again In a capitalist system, people respond to incentives and kind of monopolist like pricing power tends to not last very long. Sometimes it can last for a few years. But it never lasts forever as far as I know. What's the most extreme debt cycle that you studied in the book? And how extreme is it compared to what we have right now? And what is almost guaranteed to happen over the next 18 months?

the GFC, to lead up to the GFC, you know, because it was kind of the culmination of a debt bubble in every part of the world and in every sector. Sometimes it's governments, sometimes it's companies, sometimes it's households. In this case, actually governments weren't for the most part loving that much up. They were doing a little bit, but it wasn't too bad. But it was pretty much everywhere. you know, everybody thinks that, you know, our banks were uniquely terrible or our politicians or our government was uniquely affect us. But in reality, it was a global phenomenon and the scale of it was just kind of wild. And also, you know, one of the reasons when bonds and debt becomes particularly dangerous is when essentially it's been so long since a previous crisis that you treat it as money or money like. I mean, it's kind of one of the USPs, the ultimate selling points of bonds originally was that it was kind of a you could use it as collateral as money for certain things it was kind of because it's yeah government bonds you quite often and then over time people started using high-grade

corporate bonds like IBM or Microsoft very solid you know you can use that as collateral for loans but then of course you know in in the 2000s people started using asback securities and initially those were super solid as well and eventually you know we take things too far and you know, they were not the equivalent to money. In fact, you can trade them and some of them were close to worthless. And that I think is what really transforms almost like a humdrum market downturn or an economic recession into something nasty, like really bad, is not when you invest in something that's risky and it blows up in your face. That's fine. That's just risk and reward. They're part of it. If I invest in junk bonds, look.

If they break bad if they default, you know, I can't complain about that. Maybe I did something stupid. Maybe the company did something stupid. That's fine. Is when you invest in something you think is super safe or you base your entire kind of investment strategy or the business model of the bank around something that you thought was super safe. Proved not to be so. And that's what happened, I think, in 2008. It wasn't just the scale of the debt bubble. It was how people treated it. That's what transformed it into such a horrific. Uh financial disaster but it probably isn't my favorite crisis because there are so many to choose one like really demented ones We'll get into some some demented ones. So a principle you're saying is basically Financial crises are caused not by perceived risky assets but by assets that are perceived to be safe But that are risky you mentioned junk bonds now the more polite term, of course It is high yield and it's it's funny obviously like the the real action of risky credit lending was junk bonds when it was invented in the 1980s But now, like all of that risky lending has, a lot of it has migrated from the high yield bond market to the private credit market. So the high yield bond market is, you know, so-called safe relative to the private credit market. I'm sure the private credit people would disagree. What do you make of the rise of the private credit asset class and what have you made of the jitters in the market? And I'm curious to what degree do you think they are real versus just headlines and not much substance to them? Oh, they're real.

i've been a borderline obsessed with private credit for a long time and you know i've had many arguments with people in the industry and the nuance. I think it's a fantastic asset class i think it's fantastic idea i hope it bring grows

and grows because i actually think it does the risk the financial system that's not just marketing from. uh from from the executives in the industry I think it's great if we take basically these bundles of risks which is what every loan constitutes and that is in the investment ecosystem like in the non-bank system I think that's like a great thing but as we know like whenever people get over optimistic people do dumb shit uh people invest have invested way too much money in private credit based on very backward-looking numbers and the illusion of safety or just like frankly the lack of volatility which is just an artifice because of the lack of market-market accounting. So yeah high yield frankly it does I mean it's not safe but it's far more solid I'd say now than ever has been I mean the ratings is an obvious way to look at it like over half the market is double B now but just generally the quality even beyond ratings I think is far more solid and it throws off cash it's the technicals been great and private credit has picked up all the dicey stuff and I think that's great that's what it should be but that does mean there's been dicey stuff happening there I mean for me the the the real wake-up moment was when I was still in the United States and like I started getting cold calls and offers for private credit lines Me, I'm as a journalist. I mean, that's just astonishing, right? I mean, nobody should lend any money to any journalist ever. Oh, just to lend you money, not for you to be an investor, really. People were offering me turn loans, you know, needing you 20, 30 million dollars working capital, 30, 34 percentage points above live, but I mean, incredible space. I mean, I didn't actually take them up on it, but I just thought when we're getting to that kind of spray and pray kind of approach to origination, like there was so much money flooding into private credit. And it's a very kind of how do you find the borrowers? How do you find high quality borrowers? Well, actually in the end, you don't need to find high quality borrowers. You just need to find borrowers to take the money so you can earn fees on it. It's very similar to what we saw in 2008, not in scale, of course, but the idea that you just want to make mortgages.

Because that's how you got paid you got paid by sourcing mortgages and making them and then you know Hopefully the risk is worn by the next guy So I think in private credit too much money flooded in too quickly it has been deployed a lot of it still dry pad hasn't been deployed but it was in some cases deployed poorly and There is a default cycle that is probably going to be far worse than what the backward looking numbers look like,

you know private credit looks great if you look at the historical data, because frankly, you're not looking at the market today then. Now it's just a very different market. But that's again, that's part and parcel of finance. We want these things to happen. You want cycles, you want ups and downs, you want people to learn their lessons and they will. And hopefully at some point, private credit will dust itself off, learn from this. And the same way that the securitisation has, And come up with a a better mousetrap afterwards and and that is actually a good thing that will stick around for a long time What is the issue with the mousetrap? What and what could be better about the mousetrap? Improver credit meets the liquidity I guess and the leverage so I mean two of those two three things that will you know blow up anybody I think I Do think you can make a case that private credit could be sold to retail investors. But it has to be done exceptionally carefully and I don't believe in some semi-liquid offerings. If you're going to invest in an illiquid asset class that touts illiquidity as one of its main selling points, do not do it even in a semi-liquid format. If you invest in loans with a five-year tenor, then you should be locked up for five years.

uh because retail investors we know whatever they say whatever you know retail investors can be like people like me or people are worth you know quite a few million or even billion but people pull their money out when they're afraid and these structures aren't built for that so i hope more credit Migrates from the banking system and into the non-bank financial system, but the non-bank financial system, you know private credit firms bond funds and so on Lock up investor money for a bit longer. I don't think I mean we've built an entire financial system around the idea that like one day liquidity is some sort of God-given human right and it isn't and it shouldn't be it's actually dangerous I think even mutual funds should have, you know, my personal, you should have longer lockups. You should not be able to pull your money out daily because it actually leads to suboptimal outcomes for both you, the investor and the fund manager, because they have to make decisions knowing that, you know, money can go in and out on any given day. And private credit, that's particularly acute. So a better mass trap, you know, there are many sort of small little fiddles. I would prefer leather like investment vehicles that invest in highly levered companies not be levered themselves. So BDC zero leverage. Yeah. So zero leverage ideally. I mean, again, term leverage. If if if a BDC bought sells a

10 year bond and invests in some similar maturity assets, look, it's not ideal, but it's fine, you know.

For the public, you know, I've done done a little bit of research and a lot of it is term debt. But BDCs is just one more transparent slice of the private credit industry. But, you know, quite a lot of institutional investors that put money into private credit as the returns started falling because there was capital flushing in and they had certain return expectations. They lever up their investments in these funds. And, you know, again, done judiciously, done carefully with no recourse. Maybe that's smart, but it makes me feel uncomfortable when you basically kind of lever up an investment in a highly levered vehicle anyway So that's on the more on the institutional side I hope nobody's borrowing money from the brokerage and yellowing into BDCs, but you know teach the road and also There's a reflexive dynamic you referenced of that when money floods into an asset class it makes returns look really good So the private credit loans that were made in 2018 a ton of money flooded into 2022 to refinance those loans. So defaults were very close to zero. So even if on a fundamental basis, nothing changes, defaults will likely be higher as inflows go down, which they look like they are going to. I mean, that's very apparent in the equity market, right? Like inflows will encourage, you know, will push the asset classes up.

In BDCs, like I say, in private credit, there's a different nuance. It's not like the loan value will suddenly go to the moon suddenly because there's more money going in. But yes, it will give the private credit fund manager far more flexibility in how they manage bumps along the road, bumps along the road. But you know, only to a certain extent, there's one of the reasons why we've seen the increase in payment in kind is because some of these companies like Payment in kind is a completely viable and acceptable and important tool. In many cases, it is the right one to use for companies growing very quickly, but you know, don't want to send cash out the door right then. But I think it's unambiguous that lots of private credit funds have been using PIC as a way of deferring the pain, essentially. The danger is, of course, a lot of these companies, and this is where, for example, the default cycle comes in. It's not just the fact that the defaults have been kept probably artificially low because of the money coming in to the market, but also the recovery rates are somewhat, I'd say, fantastical rate, the

assumptions. So typically, let's say in a high yield bond, you might get 70, 80 cents on the dollar depends on where you are in the cap structure, of course, but people have penciled in I suspect unrealistic recovery rates when a lot of these companies are not gonna have any recoveries whatsoever. Let's say if they're in the software industry Where you know, there are there are no plants and factories and roads and trucks, right? It's just if the company isn't good it blows up and there's nothing there for you as a creditor So it's gonna be fascinating to watch I so I tend to be on private credit stepping back. I tend to be on those guys that I think it's it's going to be a bad default cycle. It started already, but like it's getting masked, but it's not been going to be catastrophic and the asset class deserves to survive and thrive once more, once it's been through a few of these. Definitely. And you know, in a crisis, I think some of these public BDCs are probably going to go to 30 cents or 40 cents of net asset value. And for investors with the stomach, there could be opportunity there. Yeah.

completely. I mean, buying dollars for pennies is a classic way of making a killing. The problem is when you time it, of course, and when the BDCs go further. Because people always think something can fall, can't fall any further, and it always can, unfortunately. But I'm not yoloing into BDCs go that way. Robin, everything we've talked about, the off balance sheet, debt, the hyperscalers, Nvidia, very murky, unclear what's going to look like, private credit. What themes are present in there that are present throughout the history of the rise of the bond market and in debt that you wrote about in your book? Well, in Fabulous Debt, I talked a lot about how, you know, we associate bonds with safety. And quite often that is true. But sometimes that safety can lull people into a false sense of safety and they do stupid stuff. And also bonds are just as, maybe not just as, but are also susceptible to these kind of bouts and manias that we see in the stock market most obviously. So whenever a transformative new technology comes, that typically manifests itself in both the stock and the bond market. And sometimes the most dangerous development happens in the bond market. A classic case where there were the canals and banks of the United States in the early 18th, 19th century. So after independence, the US was rebuilding itself, it was building us canals. All these states were borrowing money for banks. New York famously started with the Erie Canal.

Which was just a transformer. It was like the Apollo program of the era and it was a huge success and they saw lots of bonds and both the investors and the state made a killing out of it. But that encouraged a dub debt bubble, a bond issues bubble that ended up in half the United States being in default, like the individual states all bankrupt. And that was, you know, quite a nasty crisis that took some time. So the states, the government state governments that built the canals and they were the borrowers. Okay. Yeah, typically. So they looked at what New York had done with the Erie Canal and then borrowed a lot of money on their own balance sheets because obviously they had very little debt because the United States as a federal government had assumed all the post revolution debts. They sold lots of bonds to investors in Britain and the Netherlands and Italy and France and Germany and some of the United States. And they, you know, started banks, they built canals, they started gingerly to industrialize, but they just borrowed too much money.

There's actually a great scene in Christmas Carol by Charles Dickens, where Ebenezer Scrooge is not unfortunately in the Muppets version, which is my favorite. I watch it with my kids every Christmas. But where Ebenezer Scrooge has a nightmare and he wakes up in a cold sweat because he's had a nightmare that all his securities have been transformed into United States securities, which was Dickens' joke. about how US bonds had then by then like all been become worthless like half the states pretty much had all defaulted and some of them never repaid back the debts ever. So at the time this is in the 1840s the US was synonymous with I guess like Argentina today like a country that just defaults all the time but you know those canals were valuable. Most of the states, dust of them sounds off. And we're gonna have like quite that violent a spate of state or kind of municipal defaults in the United States since then. Same thing in the 19th century, late 19th century with the railway mania. I mean, that was just massive. I mean, if you talk about AI data centers today, you know, that's a few trillion dollars. But the equivalent, if you scale it, the size of the US economy in the 1870s and the 1890s and 1890s and so on to the present day we're talking it's the equivalent railways issued the equivalent around 10 trillion dollars of bonds it was the biggest cap explosion history and you know a lot of those railways went bust and investors quite often they get in in in England and in the Netherlands and France and Germany and Spain and Denmark

they lost their shirts But the railways were still there. And that literally like physically knapped, committed together in the United States and kind of transformed the economy, which goes to show that these manias look the very painful after the 1871 financial crash when lots of railways went bankrupt. It caused a collapse of a bank called Jay Cook, which was kind of it was the equivalent of JP Morgan going bankrupt today overnight.

It was catastrophic at the time. And it caused what was, you know, long call the Great Depression until the actual Great Depression happened. And we now call the downturn in the 1870s the Long Depression. But it's still transformed the United States because all those railways were still there. And, you know, I think it shows many years and financial crisis, although painful, sometimes. They're a good thing that like a weird thing is that the optimal number of financial crisis is arguably not zero As painful as they are to live through that's an interesting argument probably I mean, I mean basically to to to guarantee that there would be no financial crisis You'd have to have regulation speculation basically be banned and I can see I can see that definitely the downsides of that Robin I understand how someone could get into a mania about a stock. They buy the stock at 100 and it goes to 900 and they get extremely emotionally very excited. But just in terms of it, I can't wrap my head around a mania, a credit mania. I understand they exist. But like the idea of earning a sofa plus 4% on a riskier thing, it just doesn't, you know, it doesn't really get my, my, my blood pumping, you know, maybe it's not wrong with me. No, I mean, sadly, there's never been been up. There have been mean bombs.

But there aren't any meme bonds around today. I guess maybe TLT is the closest or the levered version of TLT. No, so back in the day, most bonds were actually perpetual bonds. So they were quite, you know, they lasted until the government or sometimes the company paid them back. Governments especially should perpetuate. And they were quite often sold at a discount. So they were sold at, let's say, 90 cents on the dollar. uh at an interest rate of four but then of course 90 90 or one nine nine zero nine zero yeah or 60 cents uh of the dollar or whatever right uh but there was sold at a discount which is why you could have price appreciation for the bonds as well and you know this is a different era people didn't have Bloomberg terminals it was you know quite difficult for even some smart bankers to

calculate Literally what was should be the right price for this bond. So you see bonds trade way above par We see that in modern day, but like it just shows that the bond could go up and down a lot So people could get quite excited and in an era where you know What else could you buy to make money? So let's say if you're in Change Alley, it's kind of the Wall Street of Britain in the 17th and 18th century and you're buying a bond for a newly independent latin american country well you might be buying let's say a brazilian bond this new fantastical country you've never heard of called brazil but you know the bankers are saying it's fantastically full of potential you're buying that at let's say 50 cents on the dollar well and then you're also getting the coupon maybe you're also getting the equivalent of 10 cents a dollar on interest all the time so you're getting that Plus the price keeps going up because everybody else is discovering this new country called Brazil. So that's why you can get wrapped into it. I mean, in the 19th century, there was a famous fraudster called Gregor McGregor that literally invented an entire country so he could sell a bond and he just took the money and ran to France. But you know, people didn't know better than that. Fraud is the business that has the highest profit margin. Yes, very much so. If he had to wait with it, Gregor McGregor made out like a bandit. But to be fair, like, I mean, his efforts, you know, There are frauds and then there are frauds. This guy invented an entire country. He invented a capital, a coat of arms, an entire system of government, geography had maps made, he had songs made. He just basically invented an entire country out of cloth and managed to trick hundreds of people to literally move to this country and also invest in the country's bonds. But they ended up at something called actually the mosquito coast and most of them died there, sadly.

So, you know, quite tragic end. But you know, the joke is that the difference between tragedy and comedy is time. So hopefully after 200 years, we can we can laugh at the the the buckle of poye and Gregor McGregor. Yeah, I don't think I want to go to the mosquito coast. No, it's not nice. It's somewhere in Guatemala now. But that's where he invented this country of golden honey everywhere, apparently. Robin, one thing when you mentioned that how many of the canal bonds went bad, how many of the railroad bonds went bad, but ultimately like, okay, the old man and the family made the loans and eventually the grandson was able to like recover 70 cents on the dollar because he held it. It just made me think that just

people holding the bonds like in their closet drawer and then eventually being paid back, that is much more stable than like a highly sophisticated financial institution.

Holding these securities on leverage, which is what BDCs are basically even though it is a lot of his term debt Yeah, I mean the reason why we always call like financial crisis back in the day used to be called panics because usually it was banks that held these loans these bonds and then even though a bond was you know, it's technically designed. It's supposed to be tradable Quite often when everybody wants to sell and nobody wants to buy, well, good luck trading it. And there was no deposit insurance. They had deposit of money that borrowed money themselves. So that's why banking crisis and panics, they were intertwined for a long time. Now it is different. But yes, sometimes if you buy something unlevered, you can lose money, but you can only lose what you put into it.

And that's why leverage is so so dangerous and you know has shown that again and again and again in every major and minor market cycle. So earlier you talked about the great financial crisis 2008 GFC, but you said it wasn't one of your favorites. What is one of your favorites in the book and why? God, it's like choosing my favorite child. I know it's very difficult. I mean, I do like Gregor McGregor and Poyer. I mean, it's just incredible, right? I mean, 1873, the railway mania and crash is kind of epic because it was epic in size. It was epic in its casualty. Jay Cook was, he was the John Paypoint Morgan before John Paypoint Morgan. He was Titan. He was the guy that bankrolled the North's victory in the Civil War. And suddenly he just went bankrupt out of the blue.

because he'd he'd gone over his skis on on transcontinental railway bonds. He'd he'd decided against his better judgment initially to back one of these big companies and and it just soured on him. So because the mix of like the the enormous ambition of these transcontinental railways because they weren't just like what they were like a series of Apollo programs all happening at the same time. And they did genuinely transform the United States into what we know now know today used to be kind of a coastal country. It was like North and on the eastern coast and the west coast and it went up and down But suddenly it became a country that changed its axis. It

was west to east You could actually travel from, you know, California to Maine in a few days at least or at least a week rather than months it will take before so I think the mix of both the economic impact the political Importance, you know this really did transform it kind of united the United States physically for properly for the first time and Now how nicely it ended. It was a gigantic financial crisis that we don't remember that much these days, but you know, it was it was huge almost everywhere Lots of companies went bankrupt in the United States. It's expressions like hobo came from that era because we're so many homeless people and then soldiers, unemployed soldiers also after the Civil War that lost their employment in the railway lines. So I think that's probably my favorite, but you know, I'll change. A hobo on the railroad tracks, you can't have that if there's no railroads. No, exactly. It kind of seems to me like as speculative as data-centered build that is, once the data centers are built, they are producing revenue now. It seems to me that railroads back then, we're a little bit more speculative. Like to actually literally you have to have a guy putting the the wooden tack in and then, you know, foot by foot across the entire country before the and then it has to be built before then the train has to be built. You have to market it before the revenue bill. Like that does seem to be a greater endeavor than building a data center, which is now very, very difficult and takes time and tons of capital, of course. But it seems a little easier. Yeah. I mean, don't forget, I think.

The the difference between railways in Europe and the United States is an interesting one because like in Europe railways connected existing towns like you built a railway from Liverpool to Manchester for example or from Berlin to Paris in the United States railways built towns it created entire towns it created entire states Bismarck the city is literally only main name Bismarck as a marketing gimmick for the company that built that railway line the northern pacific as a marketing gimmick to appeal to german investors because the chance of germany at the time was called autofon Bismarck and you know these were in the middle of nowhere i mean as you know even better than me i mean the united states is a vast country and back there very little of it was settled so you know it's incredibly hard work i mean obviously this is manual people had to literally hammer down the nail let to dig out the grow the rose you have to keep it smooth as well right so some backbreaking work then there are all the the ravines the mountains

the forest everything you have to go and this in the middle of nowhere it was lethal like thousands of people died As much as the data center construction is pretty epic today I am not aware of some mass casualties in the in the construction of a data center in New York yet and and and you know, this was You know the equivalent of building the pyramids essentially very epic hugely dangerous and incredibly lethal but you know transformative in the long run Yes And you're using the word epic in the British sense or the way the British people use the word great. Like it doesn't mean that it's a good thing. It just means it's big at scale. Oh, yeah, yeah. No, I mean, I think the railways are good pyramids, you know, lots of slave labor there as well. I mean, in the railways, there was, you know, a lot of it was free workers, but, you know, not always. And they were treated incredibly shabbily, especially like lots of workers were imported from China, for example. Basically.

killed in the thousands lots of Irish workers and you know it was you know a positive thing in the long run but you know not quite up to modern labor standards put that way but yes epic in the Titanic country transforming projects that unfortunately do sometimes always have a darker side as well. So if a few months ago I interviewed Leo Leaquat, Ahamed, the author of a book of 1873. A few months later, I'm listening to the Microsoft earnings call and CEO Sachin Dadella says, you know, we at the Microsoft, the executive team, we're reading an 1873, so we're thinking about this. So let's say in a few months, the next Microsoft call, you know, the team, they say, we're, we've been reading a fabulous debt epic story of how bonds built the modern world. What are some lessons that you think they should know. The people who are spending hundreds of billions of dollars, borrowing hundreds of billions of dollars, and probably, according to the off balance sheet lease commitments, it's going to be over a trillion, a trillion and a half, as you say. What are the lessons that they should know?

Well, Lyak's book is phenomenal. It's really good. You know, I take, I tackle the railway mania and the U.S. My book is a bit more U.S. centric. His is more global and focused maybe a bit more on Europe and the Grunderkrak in Europe, which is spectacular. But I hope people realize that bonds are an incredibly powerful financial technology. It's kind of the financial technology. It's kind of loans 2.0. I mean, they were both both banks and

bonds were born in Renaissance Italy a thousand years ago. But it's only now really that the bond market has, I'd argue, supplanted the banking system as the dominant credit engine of the global economy. And the reason why actually some of the basic building blocks haven't changed that much over the hundreds of years is because it's incredibly powerful.

you know, it's fixed in interest so you can calculate things easily and it's tradable and that gives you and it's decentralized. It's kind of the original to the OG decentralized finance because you can sell not just a one or two banks borrowed for a couple of a club of banks, you can sell bonds to thousands, even millions of investors indirectly. That's why you can pool individually tiny pieces of savings to one big gushing river. And that's kind of what the hyperscalers are doing. So I hope like a Microsoft or any of these CEOs and CFOs reading it would Realize that actually bonds you can iterate on the fundamental fundamental technology and people are and do but it still works and that transparency the the sobriety that comes with doing something through public fixed income markets rather than leases opaque financing arrangements private credit loans negotiated, you know off the side That comes at a cost. That's flexibility. That's great. But if you have big projects like the railways, the most valuable thing to do is to sell bonds. The bond market is supremely able to handle that and has shown that again and again and again, ranging from Napoleonic Wars, canals, railways and AI centers today. And I'd much rather that goes into the public.

Fix income markets then then then stay in the shadows Why is it in the shadows and you talk about this flexibility? You know the private property people they say oh our borrowers love flexibility I don't even really know what that actually means I agree. I mean flexibility. It just sounds great. Like yeah, you want flexibility? Yes, definitely. You want freedom? Yeah, definitely, but in practice it comes at a cost. Yeah, broadly speaking If Microsoft wants to, let's say, sell 10 billion, or wants to borrow 10 billion dollars to build a new data center, what is the cheapest way for a large mainstream public company to do so? Is it to sell to like 10 private credit firms or maybe a handful of private credit firms to do it without a rating doing quickly? We can do it opportunistically that way, yeah, sure. But you...

Definitely going to pay a lot less to borrow by just issuing a plain vanilla for general purposes corporate bond and the reason why they aren't doing that is because they want to Maybe obscure how these companies have become kind of been gone from being lean mean cash machines into being CapEx hungry utilities And maybe that pays off. I mean the returns to some of these data centers are pretty phenomenal right now But they're not doing it for purely financial reasons. And I think flexibility is probably a convenient excuse to hide, you know, that this is more about making them seem healthier than they really are. Yeah, I think one thing that like Corweave, you know, Neil Cloud is doing is delayed draw term loans. So oh, you don't have to actually borrow the money until you need it. So it's like it's like a credit line.

Yeah, I mean, the real king of debt, I would say, is Corweave that there's just reported. I've never seen a bigger gap between EBITDA and net income loss. It is quite extreme and it's a little railway like. What do you make of just, I mean, Corweave's massive, massive borrowings? Yes, it's heavily indebted. There is in every cycle one or two or maybe a handful of outliers. Look. I'm not worried about like Facebook and Alphabet or Amazon going bust. Yeah, uh, you know, they they if they take all these liabilities on balance sheet, like it's not great for investors. Uh, I worry about the financial hangover, but like it is fundamentally different in that this is not the dot com era. These companies do have solid real products and they're just shoveling all that money and a bit extra into AI. And even if AI somehow goes to zero, nothing happens. I think It's manageable. It's okay, but there will be of course in any cycle a few extreme outliers That just like borrowed way too much money did too many dumb things with what kind of to invest in this or didn't have any other products or fallbacks essentially people still gonna be going on YouTube Even if Google wastes a few hundred billion dollars on on AI data centers And that's gonna save them with a core weave or some of these other companies, do they have that backup? I mean, maybe crypto mining, I don't know, but I'd worry about those, essentially, more than I do the big hyperscalers. There are a few of the hyperscalers that look a little bit dicey here, but broadly speaking, they're probably, probably okay. Yeah, perhaps referring to Oracle, definitely the most indebted relative to its revenue.

Yeah, Oracle in the hyperscale terms is the rest and then Oracle Oracle is not like a tiny bad company or anything like that. But it's just it's not doesn't have nearly the financial and corporate heft of the others. And, you know, it's clearly the weakest of the of the litter. How long do you think this capex bubble burst? You know, I think we all know that this is not going to be infinite. She's don't go to the sky. There will be, you know, a bust, a correction like. Do you think, you know, it's going to be soon or in a few years? I mean, obviously, I have no clue. Yeah, it's like you say, trees don't grow to the sky. Capex bubbles can continue for a long time until it becomes very obviously unmanageable. Right now there are a lot of people in that industry and this is maybe both what worries me but also can keep the show going for a lot longer. There are a lot of people now with a vested interest in keeping this going.

that that AI industry has become remarkably incestuous with just an incredible tangle of financing agreements, co-investments, supplier and customer relationships that kind of bind it all together but also can kind of keep things going for a long time. They all have an interest in kind of managing this and that makes me worried about what the Daniel Moore funny looks like. Um, but it also means it can continue for a while longer. And then it, I guess, you know, but the chicken answer, it just comes down to the technology that to what extent AI genuinely is transformative. Is it glorified chatbots or is it going to cure cancer? Is it, you know, put, you know, people in Mars and, you know, where we fall on that spectrum is probably what's going to decide.

just how much those investments pay off. But the scale is pretty astonishing now. Okay, so as someone who is a journalist and is talking to people all the time and is very well informed, what are you hearing about how the revenue is at OpenAI and Anthropic? I think that literally like over half of what matters is that topic. Are you hearing good things or bad things or medium things? So I haven't spoken to anybody directly about the revenues at OpenAI and Anthropics. I only know what my colleagues have reported in the paper. I think it's broadly understood that Anthropics looks financially a lot healthier than OpenAI. And that's one of the reasons why they're probably going a little bit more aggressively for an IPO now. But I'd question with private companies, how real sometimes revenue is and not

like fending numbers, but just like, I mean, if you just look at the net income of some of the high-priced, the public companies now, look at how much is actually classified as other income, which is essentially re-evaluations of their investments. Like a lot of the money that Microsoft and Google and Amazon have made are basically marking up the value of their stakes in anthropic, open AI and SpaceX and other companies.

If you take that away then some of those earnings look ellipse a little bit Not bad, but definitely not as good. Yes, and with like Open AI and revenue how much of that is actually cash like free cash by rules everything and You know until I've seen the accounts. I don't know even when we've seen the accounts. Sometimes it's hard to know But I can tell you I'm really looking forward to the S1s for open AI and anthropic. That's gonna be a popcorn moment for me Definitely. Robin, how durable do you think the credit-raising agencies are? So Moody's, S&P, Fitch, the former two, which are publicly traded companies. And up until recently, we're viewed by the compound bros, the hedge funds as these extremely durable businesses. The valuations have fallen a lot because OAI could displace them. I just wonder, you having spent so much time researching and writing this book, just your insight on the value or lack thereof like is just just a sticker that really isn't that valuable um over time like do you think in in 2070 when there's a giant credit cycle are people going to be like oh my god i need my moody's rating before i buy it So that's a great question, actually. And because I spent a lot of time, a depressing amount of time writing a book thinking about this. There's an entire chat just on the history of the rating agencies. And it is kind of weird, like how many cow pies they've stepped in over the years and how they endure. And I think that's the secret to answering your questions that, yes, I don't know about 2070. That's a long way off. But I think people will be shocked.

at the resiliency of their business model because people don't actually pay S&P and Moody's and Fitch for their credit work. It's not like if you're the CIO of PIMCO and you sit there, well I'm going to look at what Moody says about this bond. I mean you care about the rating, the rating but you know for investment mandate reasons. But the credit work you do yourself and that's clearly like with AI, like a lot of that is happening, a lot of that happened before the current excitement about large language models. Like

I've been covering AI for, I mean, before it was cool, you know, it's a natural language processing and machine learning. I used to cover quants all the time. And it was fascinating to see how people were learning to, you know, automate the ripping apart of a credit prospectus and putting their end here and models and then automating all that. And this was 10 years ago. But the rating matters not is because of like you want Moody's to tell you what to think of this investment because they famously don't try to give investment advice. They just give a probability of default. The Moody's rate, I mean, the credit rating agency ratings are a lot better than people think. Like there are outliers when people say, well, this company was rated a half a year ago and went bust, but they are highlighted because they're actually pretty rare. Broadly speaking, the letter base model as a signifier of the chance of default is actually pretty accurate. Even the financial crisis, all those shoddy, securitized monstrosities that were given AAA ratings. Well, actually, even quite a lot of the AAA tranches ended up being money good. They traded down maybe to 20 cents on the dollar, but a lot of those actually were pretty okay.

And AAA companies, AAA governments, you know, there aren't that many of them around these days, but it tends to work. And I think the reason why the rating agencies actually endure will continue to endure is because this phrase I once came across somebody in the industry use, but you talked about the need for a language of credit. Like we need shorthand. We're humans. We're both very smart and very stupid at the same time, and we like these shorthands. We like rules of thumb. We like simplistic models. And it's just nice to have something. This is a single B. That's a double A. That's a triple C. And the reason why the rating agencies, despite having very different, they talk up all their difference, they still have basically the same letters as well. And that's because it gives us a cohesive language to talk about credit.

And sometimes it's wrong, like all language can be, all models don't work. You know, it's like, there's a famous British statistician who said that all models are wrong, but some are useful. The rating agency models are not as wrong as people think, and it's still pretty useful. And as much as you can automate all sorts of cool shit with AI, I think that will endure. And the craving for just a brand, a name, like a Moody's or an S&P. Is is going to

stay there and in fact in places like the United States is enshrined in law Despite all the controversy around the financial crisis, you know, the nationally They recognize rating agency designation is is still there. It's still in the books and that's why You know, it's kind of one of the most stubborn oligopolies in the history of business probably And if you're an insurance company buying something, you have to buy something. A certain percentage of your assets have to be investment grade, even if the rating is totally wrong. And also, I think of the Charlie Munger anecdote about how he was in World War II, and I think he was tracking the weather. And he also was starting to be superior, like, hey, my forecasts are really bad. You shouldn't be asking me for these forecasts. And the military people said, We need these forecasts for our military planning. So even though the forecasts are wrong, we still need them. We still need them. We still need something like that. Yeah. I mean, it's like so many things in finance that look weird. I mean, in the world, really, they look weird or dumb or dangerous. It's quite often like you still come to this kind of, well, if it didn't exist, we'd have to invent it. Ratings as weird and dumb as they sometimes can seem, we still need something like that. And if they didn't exist, we'd have to reinvent them all over again.

The insurance issue is quite interesting drawing back to private credit there, of course There is always a danger of shopping around for the greatest Rating and broadly speaking that the big three have done a pretty good job over time not always but over time to as much as you know, they could be a little bit more Commercial let's say it's certainly before the financial crisis and broadly speaking not letting the standards arose too comically far But clearly there, I do worry about so-called private label credits that insurance companies are getting on private credit loans and saving their investment grade when really the reality is, I suspect, a lot iffy. What does an investment grade private loan really, really mean?

What it means you can you literally have something you can put internally in your model saying oh this this By that this direct loan to widget maker XYZ or acne software is a single-way credit and therefore the amount of capital or how much you have to plug in your models of risk and default on that is lower and Ergo, you know your return to your investment returns look better I think most of the big series Insurance companies are very

aware of this issue and are aware of it. And if they do use private label, a label credit ratings, they take it with a pinch of salt and they know the issues, they'd be dragons maybe. But there are also a lot of private insurance companies that are owned by private equity. And those private equity insurers, companies sometimes own also some of these private label companies. And I do worry about that.

the tangled private capital ecosystem of private credit, private equity, private ratings, and private equity owned insurance companies. I think that is something that could at some point bear watching as well. Have you looked into these things called funding agreement backed notes? No, but it sounds amazing. Tell me more. It is basically when an insurance company, probably like a life insurance company, Issues debt, but the debt that they're issuing they can call it a policy a life life insurance policy. Yeah Yes, no, actually, I do remember reading about this and I was delighted to learn about it It shows that there is nothing more creative on this planet as a financial engineer who wants to optimize risk reward and game the system to do so Is it you know One of the the dangers of my job journalism and your job and I guess everybody's job is that we we look very much backwards and It's always cooler to seem pessimistic and then call this is the next big thing and this is the next CDO or whatever and You know luckily those kind of crashes like 2008 they don't happen very often I actually have like literally in front of my desk I have a little cartoon that shows it's from 2008 that shows like somebody going up in the airline and the captain comes across the town always saying oh you know there's a bit of turbulence buckle up and there's a passenger who screams oh my god we're all going to die and the passenger next to him says look just it's a financial journalist don't worry he's just panicking And I just have it there in front of my desk to remind myself and not always thinking everything is the next 2008. Not everything is a big crisis. So these these notes, look, I think it's the optics are bad. The fundamentals are probably not great. Is it going to be a disaster? Probably not because, you know, yeah, it might be bad. But, you know, we can't have reward without risk. You can't make money without losing money. That's kind of what keeps the train on the roads. And you know people sometimes create stupid things invent new things game the rules and they get their faces ripped off it blows up in the next downturn but the good dimensions they survive and they evolve and they thrive and

securitization is one of them securitization was a dirty word not that long ago. Like I'm just a few years ago and now we're looking even the Europeans are talking about it like oh my god we wish we had America's mortgage backed security market. Oh my god that would be amazing. And I remember when even American politicians were badmouthing it. Stupid things happened in 2008 but we learned from it. And you know at some point we'll realize what was really stupid that we're doing right now, what was actually just fine and what was just moderately stupid. And then I get to write a book about it a few years after that. So, you know, it's all, it's all gravy for me as a financial journalist. People should buy the book, a fabulous debt, the epic story of how bonds built the modern world. Buy it for yourself, buy it for your kid, buy it for your parents, grandparents. Thanks so much. Thanks for having me on, Jack.